



Cloud Application Developer

QP Code: SSC/Q8303

Version: 4.0

NSQF Level: 5

IT-ITeS SSC NASSCOM || NASSCOM Plot No - 7, 8, 9 & 10, 3rd Floor, Sector 126
Noida Uttar Pradesh - 201303 || email:sscstandards@nasscom.in

Qualification Pack

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SSC/Q8303: Cloud Application Developer

Brief Job Description

A Cloud Application Developer is responsible for designing, developing, and maintaining secure, resilient, and scalable cloud-native applications. The person gathers and validates functional and non-functional requirements, implements modern architectures and DevSecOps practices, and manages the full application lifecycle using continuous integration and delivery (CI/CD) pipelines. The individual also ensures regulatory compliance, application performance, and efficient use of cloud infrastructure and services.

Personal Attributes

Cloud Application Developers must be able to communicate effectively, build relationships with others and collaborate with various stakeholders. They must have a zeal to learn new things.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8323: Monitor and manage cloud applications and the deployed systems](#)
2. [SSC/N8322: Package software for secure and successful deployment on the cloud](#)
3. [SSC/N8321: Migrate applications to utilize the full potential of the cloud platform](#)
4. [SSC/N8320: Develop and maintain secure, resilient and highly available cloud applications](#)
5. [SSC/N8319: Design the application architecture to ensure scalable, resilient, and fault-tolerant cloud applications](#)
6. [SSC/N8318: Develop functional and non-functional requirements for the defined scope of the application](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing

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Country	India
NSQF Level	5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2512.0204
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (UG in Computer Science/ Information Technology/ Electrical and Electronics Engineering. Diploma in CS/IT/Electronics) OR Completed 3 year diploma after 10th (Diploma in CS/IT/Electronics) with 1.5 years of experience Relevant Experience in cloud application development or DevOps OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience Relevant Experience in cloud application development or DevOps
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	4.0
Reference code on NQR	QG-05-IT-045902025-V2-NASSCO
NQR Version	2

Qualification Pack

SSC/N8323: Monitor and manage cloud applications and the deployed systems

Description

This unit is about monitoring and managing the performance of the applications and the deployed systems.

Scope

The scope covers the following :

- Gather requirements
- Monitor system performance
- Report application performance

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify performance-impacting factors of the cloud application, such as memory leaks, traffic loads, or latency
- PC2.** evaluate application performance bottlenecks using historical performance data and logs
- PC3.** define and document cost, resource utilization, and scalability requirements for optimization
- PC4.** define clear metrics (e.g., latency, error rate, CPU/memory usage) to track system health
- PC5.** configure dashboards and alerts using monitoring tools like CloudWatch, Datadog, or New Relic
- PC6.** use AI/ML-powered tools to detect anomalies and predict potential system failures
- PC7.** perform root-cause analysis based on error logs and AI-based diagnostics
- PC8.** validate cloud resource utilization against cost benchmarks and SLA parameters
- PC9.** implement automation scripts to respond to predefined thresholds or anomalies
- PC10.** monitor applications for security breaches and ensure compliance with cloud security policies
- PC11.** create structured reports with visualizations to summarize performance data and trends
- PC12.** share performance reports with stakeholders using approved templates or collaboration tools
- PC13.** provide actionable recommendations for improving performance, cost-efficiency, and scalability
- PC14.** document any anomalies detected by AI/ML tools and suggest remedial steps

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies and standards for monitoring and managing deployed systems
- KU2.** procedures for reporting performance metrics to relevant stakeholders
- KU3.** templates and tools used for documenting and sharing performance insights

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- KU4.** factors affecting the performance and cost-efficiency of cloud applications
- KU5.** metrics used to monitor application health, availability, and performance
- KU6.** tools and platforms available for cloud application monitoring and how to configure them
- KU7.** principles of cloud resource optimization and workload balancing
- KU8.** capabilities of AI/ML-based tools for performance monitoring and anomaly detection
- KU9.** techniques for automating responses to performance thresholds or anomalies
- KU10.** common security threats in cloud applications and practices for secure monitoring
- KU11.** data privacy and compliance considerations in application monitoring
- KU12.** reporting formats, visualization techniques, and tools used to communicate performance trends

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate clearly and confidently with team members and stakeholders
- GS2.** interpret and analyze technical documents, logs, and performance reports
- GS3.** document findings, analysis, and recommendations in a structured and concise manner
- GS4.** solve problems independently by identifying the root cause and implementing appropriate solutions
- GS5.** make timely decisions under pressure when monitoring anomalies or system alerts
- GS6.** collaborate effectively with cross-functional teams to resolve system issues
- GS7.** prioritize tasks and manage time efficiently to meet monitoring and reporting deadlines
- GS8.** demonstrate attention to detail while interpreting performance data or writing reports
- GS9.** adapt to new tools and technologies relevant to cloud monitoring and AI/ML integration
- GS10.** maintain composure and professionalism when handling system issues in a live environment

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. identify performance-impacting factors of the cloud application, such as memory leaks, traffic loads, or latency	2	3	-	1
PC2. evaluate application performance bottlenecks using historical performance data and logs	2	3	-	2
PC3. define and document cost, resource utilization, and scalability requirements for optimization	2	4	-	1
PC4. define clear metrics (e.g., latency, error rate, CPU/memory usage) to track system health	2	3	-	2
PC5. configure dashboards and alerts using monitoring tools like CloudWatch, Datadog, or New Relic	2	4	-	2
PC6. use AI/ML-powered tools to detect anomalies and predict potential system failures	2	4	-	2
PC7. perform root-cause analysis based on error logs and AI-based diagnostics	2	4	-	1
PC8. validate cloud resource utilization against cost benchmarks and SLA parameters	2	3	-	1
PC9. implement automation scripts to respond to predefined thresholds or anomalies	2	4	-	1
PC10. monitor applications for security breaches and ensure compliance with cloud security policies	2	4	-	1
PC11. create structured reports with visualizations to summarize performance data and trends	2	4	-	2
PC12. share performance reports with stakeholders using approved templates or collaboration tools	2	3	-	1
PC13. provide actionable recommendations for improving performance, cost-efficiency, and scalability	3	4	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. document any anomalies detected by AI/ML tools and suggest remedial steps	3	3	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8323
NOS Name	Monitor and manage cloud applications and the deployed systems
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	5
Credits	2
Version	4.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8322: Package software for secure and successful deployment on the cloud

Description

This unit is about building, packaging, and deploying cloud-native applications using automated, secure, and intelligent approaches. It includes incorporating modern technologies such as containerization, AI/ML-based optimization, and automated pipelines for robust deployment.

Scope

The scope covers the following :

- Build & package cloud applications
- Plan for deployment
- Automate deployment

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** create deployment units from application source code using appropriate build tools
- PC2.** configure and manage environment-specific service configurations for deployment
- PC3.** apply secure coding and deployment practices to protect application code and configurations
- PC4.** use suitable containerization technologies (e.g., Docker) to ensure environment-agnostic deployment
- PC5.** incorporate AI/ML-based code analysis tools to detect vulnerabilities or inefficiencies during the packaging phase
- PC6.** define deployment strategies based on application architecture, performance, and downtime requirements
- PC7.** develop and maintain an operational checklist covering procedures, risks, and rollback strategies
- PC8.** evaluate the use of AI/ML models to forecast deployment risks and recommend optimal deployment windows
- PC9.** automate the deployment process using CI/CD pipelines, infrastructure-as-code (IaC), and scripting tools
- PC10.** plan and configure container orchestration platforms (e.g., Kubernetes) to provision, deploy, and manage services
- PC11.** integrate AI/ML tools for real-time deployment monitoring and anomaly detection
- PC12.** coordinate with infrastructure, QA, and support teams to resolve pre- and post-deployment issues

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** policies, procedures, and best practices for cloud application deployment
- KU2.** data protection and sharing protocols during packaging and deployment
- KU3.** guidelines for documenting deployment strategies and operational readiness
- KU4.** roles and responsibilities of team members involved in the deployment process
- KU5.** standard CI/CD tools and templates for automating builds and deployments
- KU6.** process of creating, configuring, and validating deployment units
- KU7.** principles of secure configuration and credential management
- KU8.** containerization concepts and tools (e.g., Docker, Podman)
- KU9.** container orchestration platforms (e.g., Kubernetes, OpenShift)
- KU10.** use of AI/ML for automating and optimizing deployments (e.g., anomaly detection, auto-healing, predictive scaling)
- KU11.** types of deployment strategies (blue-green, rolling, canary, etc.)
- KU12.** security standards such as OWASP, DevSecOps practices, and compliance norms
- KU13.** version control systems and dependency management practices
- KU14.** operational metrics to monitor during deployment and post-deployment phases

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate technical information clearly and accurately to team members and stakeholders
- GS2.** interpret and follow documentation, code comments, and deployment guidelines
- GS3.** work independently and manage tasks within collaborative teams
- GS4.** analyze situations and make informed decisions under tight timelines
- GS5.** identify and solve technical issues quickly and efficiently
- GS6.** coordinate with cross-functional teams to ensure a smooth deployment process
- GS7.** plan and prioritize work effectively to meet release deadlines
- GS8.** document deployment processes and maintain logs for future reference
- GS9.** evaluate risks and follow contingency plans where needed
- GS10.** adapt to new technologies and tools relevant to cloud deployment and automation

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. create deployment units from application source code using appropriate build tools	2	4	-	1
PC2. configure and manage environment-specific service configurations for deployment	2	3	-	2
PC3. apply secure coding and deployment practices to protect application code and configurations	2	3	-	1
PC4. use suitable containerization technologies (e.g., Docker) to ensure environment-agnostic deployment	2	4	-	1
PC5. incorporate AI/ML-based code analysis tools to detect vulnerabilities or inefficiencies during the packaging phase	2	4	-	1
PC6. define deployment strategies based on application architecture, performance, and downtime requirements	4	5	-	2
PC7. develop and maintain an operational checklist covering procedures, risks, and rollback strategies	3	5	-	2
PC8. evaluate the use of AI/ML models to forecast deployment risks and recommend optimal deployment windows	3	4	-	2
PC9. automate the deployment process using CI/CD pipelines, infrastructure-as-code (IaC), and scripting tools	2	5	-	2
PC10. plan and configure container orchestration platforms (e.g., Kubernetes) to provision, deploy, and manage services	2	5	-	2
PC11. integrate AI/ML tools for real-time deployment monitoring and anomaly detection	3	4	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. coordinate with infrastructure, QA, and support teams to resolve pre- and post-deployment issues	3	4	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8322
NOS Name	Package software for secure and successful deployment on the cloud
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	5
Credits	3
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8321: Migrate applications to utilize the full potential of the cloud platform

Description

This unit is about migrating applications to the cloud by incorporating changes in the application code, applying AI/ML-enabled tools and methods, and ensuring the cloud platform is utilized to its full potential in terms of scalability, automation, performance, and cost-efficiency.

Scope

The scope covers the following :

- Gather business objectives
- Analyze the source code
- Map cloud and on-premise data
- Re-host applications
- Re-platform OS/middleware
- Re-factor/ reengineer/re-architect applications
- Test the applications
- Ensure smooth operations

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify business goals and performance requirements using AI-driven assessment tools and analytics dashboards
- PC2.** identify AI/ML-based optimization opportunities to reduce cost and enhance performance in cloud migration
- PC3.** analyze source code and architecture using automated static/dynamic analysis tools, including AI-powered code scanners for migration compatibility
- PC4.** perform intelligent data mapping using AI-enabled tools for schema mapping, transformation logic, and minimizing data loss
- PC5.** identify application components suitable for lift-and-shift using automated cloud migration platforms
- PC6.** migrate compatible applications with minimal changes, supported by orchestration tools and AI-based dependency resolution
- PC7.** analyze OS/middleware compatibility and recommend re-platforming options using AI/ML-based migration assistance tools
- PC8.** reengineer non-compatible components using AI-assisted code conversion and optimization tools
- PC9.** refactor application code using ML-powered performance profiling and automated remediation tools
- PC10.** incorporate cloud-native and AI-first architecture principles for scalability, security, and resilience

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- PC11.** develop API-based integrations for hybrid data migration, utilizing AI for integration testing and dependency mapping
- PC12.** implement AI/ML-driven testing frameworks to validate performance, reliability, and scalability
- PC13.** conduct predictive unit testing using AI-enabled test case generation and anomaly detection tools
- PC14.** implement intelligent SOPs and AI-based monitoring dashboards for proactive management of cloud operations post-migration

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** cloud migration and data governance procedures
- KU2.** data sharing and retention protocols including privacy and compliance aspects
- KU3.** procedures for documenting migration plans, SOPs, and test protocols
- KU4.** stakeholders and roles in the migration planning process
- KU5.** standard templates, AI/ML-based migration tools, and automation platforms
- KU6.** types of application and infrastructure dependencies
- KU7.** cloud deployment models (IaaS, PaaS, SaaS, hybrid, multi-cloud)
- KU8.** cloud-native frameworks and AI/ML integration patterns
- KU9.** automated source code analysis, AI-assisted code modernization, and cloud compatibility evaluation
- KU10.** methods for intelligent data mapping and synchronization
- KU11.** lift-and-shift (re-host), containerization, and platform reengineering
- KU12.** AI/ML-powered re-factoring tools and microservices-based architecture
- KU13.** cloud security, governance, and AI-integrated DevSecOps practices
- KU14.** test frameworks using ML for test generation, defect prediction, and optimization
- KU15.** AI/ML-based observability and monitoring tools for continuous operations
- KU16.** best practices for leveraging AI and ML in each phase of cloud migration

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow AI-assisted migration plans, guidelines, and tools effectively
- GS2.** work collaboratively in agile teams and independently on assigned tasks
- GS3.** verify correctness of code, scripts, and configurations using AI validation tools
- GS4.** assess the architectural implications of AI integration during migration
- GS5.** communicate technical and operational information with stakeholders
- GS6.** apply critical thinking and informed judgment in complex cloud-AI environments
- GS7.** ensure adherence to customer SLAs and compliance benchmarks
- GS8.** maintain consistent and adaptive service delivery

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GS9. plan, organize, and prioritize tasks effectively

GS10. resolve technical challenges with AI-enhanced troubleshooting and root cause analysis tools

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. identify business goals and performance requirements using AI-driven assessment tools and analytics dashboards	2	4	-	1
PC2. identify AI/ML-based optimization opportunities to reduce cost and enhance performance in cloud migration	2	3	-	1
PC3. analyze source code and architecture using automated static/dynamic analysis tools, including AI-powered code scanners for migration compatibility	2	3	-	1
PC4. perform intelligent data mapping using AI-enabled tools for schema mapping, transformation logic, and minimizing data loss	2	3	-	1
PC5. identify application components suitable for lift-and-shift using automated cloud migration platforms	2	3	-	2
PC6. migrate compatible applications with minimal changes, supported by orchestration tools and AI-based dependency resolution	2	4	-	2
PC7. analyze OS/middleware compatibility and recommend re-platforming options using AI/ML-based migration assistance tools	3	3	-	1
PC8. reengineer non-compatible components using AI-assisted code conversion and optimization tools	2	4	-	2
PC9. refactor application code using ML-powered performance profiling and automated remediation tools	2	4	-	2
PC10. incorporate cloud-native and AI-first architecture principles for scalability, security, and resilience	2	4	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. develop API-based integrations for hybrid data migration, utilizing AI for integration testing and dependency mapping	2	4	-	2
PC12. implement AI/ML-driven testing frameworks to validate performance, reliability, and scalability	2	4	-	1
PC13. conduct predictive unit testing using AI-enabled test case generation and anomaly detection tools	2	3	-	1
PC14. implement intelligent SOPs and AI-based monitoring dashboards for proactive management of cloud operations post-migration	3	4	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8321
NOS Name	Migrate applications to utilize the full potential of the cloud platform
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	5
Credits	2
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8320: Develop and maintain secure, resilient and highly available cloud applications

Description

This unit is about building cloud applications that are secure, resilient to failure and are highly available to customers.

Scope

The scope covers the following :

- Develop cloud applications
- Ensure application availability
- Ensure application security

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify and map service dependencies (e.g., databases, message brokers) for deployment in distributed environments like Kubernetes
- PC2.** design and build resilient applications using fault-tolerant patterns such as retries, circuit breakers, and fallback mechanisms for automated failover and recovery
- PC3.** ensure horizontal scalability through container orchestration, auto-scaling policies, and effective load balancing
- PC4.** implement microservices architectures and use API gateways and service meshes (e.g., Istio) for secure, modular, and scalable services
- PC5.** ensure applications are operational, functional, and usable for fulfilling business or users' requirements by incorporating AI-driven monitoring and alerting tools (e.g., AIOps platforms) to detect anomalies and preemptively resolve downtime risks
- PC6.** integrate DevSecOps practices to implement and maintain security controls in compliance with standards (e.g., ISO 27001, GDPR)
- PC7.** secure data in transit using HTTPS, TLS, or VPNs, and define network security policies
- PC8.** implement identity and access management using protocols such as OAuth 2.0, OpenID Connect, or SAML, and enforce role-based access control
- PC9.** secure data at rest using encryption techniques (e.g., AES, RSA) and cloud-native key management solutions (e.g., AWS KMS, Azure Key Vault)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** industry standards for secure cloud application development and deployment
- KU2.** legal and regulatory frameworks applicable to cloud security and data protection
- KU3.** best practices for data integration across distributed systems

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- KU4.** process of managing application dependencies through service discovery and dependency injection
- KU5.** process of designing stateless applications with horizontal scalability

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** document and communicate technical concepts clearly to both technical and non technical stakeholders
- GS2.** seek expert advice or mentoring when dealing with unfamiliar technologies or complex situations
- GS3.** systematically debug and resolve issues using analytical tools and techniques
- GS4.** assess the implications of architectural choices on performance availability and security
- GS5.** review source code infrastructure configurations and system designs for compliance and quality assurance

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. identify and map service dependencies (e.g., databases, message brokers) for deployment in distributed environments like Kubernetes	4	6	-	2
PC2. design and build resilient applications using fault-tolerant patterns such as retries, circuit breakers, and fallback mechanisms for automated failover and recovery	3	6	-	2
PC3. ensure horizontal scalability through container orchestration, auto-scaling policies, and effective load balancing	3	5	-	2
PC4. implement microservices architectures and use API gateways and service meshes (e.g., Istio) for secure, modular, and scalable services	3	5	-	2
PC5. ensure applications are operational, functional, and usable for fulfilling business or users' requirements by incorporating AI-driven monitoring and alerting tools (e.g., AIOps platforms) to detect anomalies and preemptively resolve downtime risks	3	6	-	2
PC6. integrate DevSecOps practices to implement and maintain security controls in compliance with standards (e.g., ISO 27001, GDPR)	5	5	-	2
PC7. secure data in transit using HTTPS, TLS, or VPNs, and define network security policies	5	5	-	3
PC8. implement identity and access management using protocols such as OAuth 2.0, OpenID Connect, or SAML, and enforce role-based access control	2	6	-	3
PC9. secure data at rest using encryption techniques (e.g., AES, RSA) and cloud-native key management solutions (e.g., AWS KMS, Azure Key Vault)	2	6	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8320
NOS Name	Develop and maintain secure, resilient and highly available cloud applications
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	5
Credits	3
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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SSC/N8319: Design the application architecture to ensure scalable, resilient, and fault-tolerant cloud applications

Description

This unit is about designing scalable application architecture which is secure, reliable, and resilient to failure.

Scope

The scope covers the following :

- Design cloud application architecture
- Secure the cloud application

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify appropriate cloud deployment and service models (e.g., public, private, hybrid, IaaS, PaaS, FaaS) based on scalability, availability, and workload requirements
- PC2.** apply cloud-native architecture patterns using frameworks and guidelines (e.g., twelve-factor apps, well-architected frameworks) to design modular and scalable solutions
- PC3.** perform impact analysis of architectural decisions using trade-off models (e.g., cost-performance-latency trade-offs) and tools like AWS Well-Architected Tool or Azure Advisor
- PC4.** implement modern application design paradigms such as serverless, container-based microservices, and event-driven design using technologies like Kubernetes, Lambda, or Azure Functions
- PC5.** define and document infrastructure as code (IaC) specifications for reliable provisioning using tools such as Terraform or AWS CloudFormation
- PC6.** design APIs and reusable services using RESTful or GraphQL architectures, and manage them using API gateways and service meshes
- PC7.** identify components requiring high availability and apply appropriate deployment strategies (e.g., active-active, auto-scaling, multi-zone replication)
- PC8.** incorporate security-by-design principles such as Zero Trust Architecture, DevSecOps, and secure SDLC practices in application architecture
- PC9.** identify potential attack vectors, vulnerabilities, and threats using automated threat modeling tools and AI-powered risk analysis
- PC10.** define and document recovery strategies including backup automation, self-healing mechanisms, and disaster recovery as code (DRaaS)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** cloud deployment and service models and selection criteria based on application requirements

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- KU2.** API development and management using REST GraphQL service meshes and gateway solutions
- KU3.** scalability and resiliency design patterns such as circuit breakers retry mechanisms bulkheads and event sourcing
- KU4.** concepts of DevSecOps pipelines including automated security scanning and compliance as code
- KU5.** disaster recovery planning high availability deployments and geographic failover configurations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow and apply emerging trends in AI assisted architecture validation cloud native tools and compliance automation
- GS2.** apply structured problem solving methods to identify performance or design issues early
- GS3.** manage competing priorities and adapt design decisions to evolving technical and business needs
- GS4.** use analytical reasoning and decision matrices to compare architectural trade offs
- GS5.** work collaboratively with cross functional teams including security platform engineering and business stakeholders

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. identify appropriate cloud deployment and service models (e.g., public, private, hybrid, IaaS, PaaS, FaaS) based on scalability, availability, and workload requirements	3	5	-	2
PC2. apply cloud-native architecture patterns using frameworks and guidelines (e.g., twelve-factor apps, well-architected frameworks) to design modular and scalable solutions	3	5	-	2
PC3. perform impact analysis of architectural decisions using trade-off models (e.g., cost-performance-latency trade-offs) and tools like AWS Well-Architected Tool or Azure Advisor	3	5	-	2
PC4. implement modern application design paradigms such as serverless, container-based microservices, and event-driven design using technologies like Kubernetes, Lambda, or Azure Functions	3	5	-	2
PC5. define and document infrastructure as code (IaC) specifications for reliable provisioning using tools such as Terraform or AWS CloudFormation	3	5	-	2
PC6. design APIs and reusable services using RESTful or GraphQL architectures, and manage them using API gateways and service meshes	3	5	-	2
PC7. identify components requiring high availability and apply appropriate deployment strategies (e.g., active-active, auto-scaling, multi-zone replication)	3	5	-	2
PC8. incorporate security-by-design principles such as Zero Trust Architecture, DevSecOps, and secure SDLC practices in application architecture	3	5	-	2
PC9. identify potential attack vectors, vulnerabilities, and threats using automated threat modeling tools and AI-powered risk analysis	3	5	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. define and document recovery strategies including backup automation, self-healing mechanisms, and disaster recovery as code (DRaaS)	3	5	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8319
NOS Name	Design the application architecture to ensure scalable, resilient, and fault-tolerant cloud applications
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	5
Credits	3
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8318: Develop functional and non-functional requirements for the defined scope of the application

Description

This unit is about identifying, documenting, and validating functional and non-functional requirements for a cloud application to align with stakeholder needs, business goals, and regulatory standards. It includes developing user stories, technical specifications, and quality attributes that guide the design of scalable and secure applications.

Scope

The scope covers the following :

- Gather functional requirements
- Gather non-functional requirements
- Validate requirements

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify and document functional requirements based on the defined scope of the cloud application and stakeholder expectations
- PC2.** analyze and describe business processes, user roles, and critical workflows to support functional mapping
- PC3.** create user stories or use cases to represent interactions between users, services, and the cloud application
- PC4.** define usability expectations, UI/UX needs, and data model requirements relevant to the cloud application
- PC5.** define system boundaries and external interfaces for integration with third-party or legacy systems
- PC6.** identify performance, capacity, and scalability expectations (e.g., latency, throughput, concurrent users)
- PC7.** document target operating environments, including platforms, devices, and network conditions
- PC8.** specify non-functional needs related to maintainability, portability, interoperability, and disaster recovery
- PC9.** establish service level agreements (SLAs) and uptime/reliability objectives for the cloud application
- PC10.** identify compliance, data protection, and regulatory requirements applicable to the system
- PC11.** consolidate and communicate documented requirements for review with stakeholders and technical teams
- PC12.** validate requirements through walkthroughs, peer reviews, or prototyping sessions to ensure completeness, clarity, and feasibility

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** procedures, templates, and tools for documenting system requirements
- KU2.** policies and guidelines for data sharing, privacy, and stakeholder engagement
- KU3.** roles of stakeholders involved in requirements elicitation and validation
- KU4.** techniques for documenting business processes using flowcharts or BPMN
- KU5.** methods to translate user needs into user stories, personas, or use cases
- KU6.** basics of user experience (UX) design, wireframes, and visual design standards
- KU7.** the principles of data modeling and commonly used database systems (SQL/NoSQL)
- KU8.** the characteristics of cloud-native frameworks and microservices architectures
- KU9.** non-functional requirement categories such as performance, security, reliability, and compliance
- KU10.** load balancing strategies and how they impact performance and scalability
- KU11.** concepts of SLAs, SLOs, and uptime guarantees in cloud environments
- KU12.** major compliance and data governance standards (e.g., GDPR, HIPAA, ISO/IEC 27001)

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively with technical and non-technical stakeholders to gather and validate requirements
- GS2.** listen actively and seek clarification to ensure accurate understanding of user needs
- GS3.** collaborate with cross-functional teams including business analysts, architects, and product managers
- GS4.** analyze how functional and non-functional requirements impact system architecture and design choices
- GS5.** ensure accuracy and completeness of requirement documentation using industry best practices
- GS6.** plan, prioritize, and manage tasks to meet project deadlines and stakeholder expectations
- GS7.** apply analytical thinking to resolve ambiguities and conflicting requirements
- GS8.** review and refine requirements iteratively through user feedback and prototyping

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. identify and document functional requirements based on the defined scope of the cloud application and stakeholder expectations	3	5	-	3
PC2. analyze and describe business processes, user roles, and critical workflows to support functional mapping	3	4	-	1
PC3. create user stories or use cases to represent interactions between users, services, and the cloud application	2	4	-	1
PC4. define usability expectations, UI/UX needs, and data model requirements relevant to the cloud application	2	4	-	1
PC5. define system boundaries and external interfaces for integration with third-party or legacy systems	2	5	-	2
PC6. identify performance, capacity, and scalability expectations (e.g., latency, throughput, concurrent users)	2	5	-	2
PC7. document target operating environments, including platforms, devices, and network conditions	2	5	-	1
PC8. specify non-functional needs related to maintainability, portability, interoperability, and disaster recovery	2	3	-	1
PC9. establish service level agreements (SLAs) and uptime/reliability objectives for the cloud application	2	2	-	1
PC10. identify compliance, data protection, and regulatory requirements applicable to the system	2	1	-	1
PC11. consolidate and communicate documented requirements for review with stakeholders and technical teams	4	6	-	3

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. validate requirements through walkthroughs, peer reviews, or prototyping sessions to ensure completeness, clarity, and feasibility	4	6	-	3
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8318
NOS Name	Develop functional and non-functional requirements for the defined scope of the application
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	5
Credits	2
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8323.Monitor and manage cloud applications and the deployed systems	30	50	0	20	100	20
SSC/N8322.Package software for secure and successful deployment on the cloud	30	50	0	20	100	15
SSC/N8321.Migrate applications to utilize the full potential of the cloud platform	30	50	0	20	100	15
SSC/N8320.Develop and maintain secure, resilient and highly available cloud applications	30	50	0	20	100	15
SSC/N8319.Design the application architecture to ensure scalable, resilient, and fault-tolerant cloud applications	30	50	0	20	100	15
SSC/N8318.Develop functional and non-functional requirements for the defined scope of the application	30	50	0	20	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	200	330	-	120	650	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.